

Introduction to AI

Working Group Session 4

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Whatsapp group for first year BSc AI students...

inappropriate messages...

The university expects everyone involved in the university community as an employee, student or otherwise to deal with each other respectfully, manifested in the style and tone of communication, both face to face and online.

If you feel uncomfortable, please consult
<https://vuweb.vu.nl/en/about-vu/more-about/safe-social-setting-on-campus> for more information and help.

If you feel personally insulted, please contact your study advisor.

Announcements

1. **Social event** (together with the programming teachers)

Register via:

<https://docs.google.com/presentation/d/1mg6ZDQDDuB6l0lYqYggPj6Zmn7vDhTlNfS1WcTMhpn0/edit?usp=sharing>

2. **Study career sessions**

First year students will have to follow 3 Study and Career sessions in period 2, and 3 sessions in period 3 to fulfill the Study and Career Learning objectives that incorporated in the course Bachelor Project Artificial Intelligence XB_40002. Submitting the assignment and the reflection for the Study and Career sessions is a requirement for the BSc project.

- **15 November 2022 15:30**
- **22 November 2022 15:30**
- **13 December 2022 15:30**

TODAY

DEADLINE	Assignment	7-sep.	14-sep.	21-sep.	28-sep.	5-okt.	12-okt.	19-okt.
16-sep.	Research problem	working on	presentation					
23-sep.	Literature		working on	presentation				
30-sep.	Approach			working on	presentation			
14-okt.	Exp. evaluation				working on	presentation		
21-okt.	Poster					working on	present	
21-okt.	Video						working on	
28-okt.	Final report							working on

Assignment: APPROACH

- Design a computational solution (“approach”) for your task. Describe the (sub)steps of your approach.
- We expect at least:
 - > a flowchart of your system
 - > a description of the chosen AI method to solve your problem
 - > a motivation for this
- Today: Presentations!
- Submit Friday 30 Sept

Assignment: Approach

- Flowchart: a type of diagram that represents a process.

A flowchart can be used as a diagrammatic ***representation of an algorithm***, a step-by-step approach to solving a task.

- Flowcharts:
 - > steps as different type of boxes
 - > ordering of steps by arrows

Assignment: Approach (Flowchart)



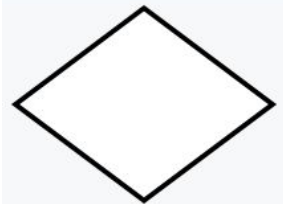
Flowline



Terminal (ending of program or subprogram)



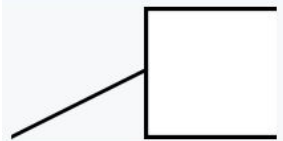
Process



Decision



Input/Output

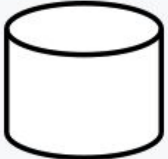
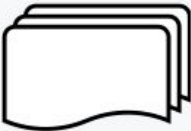
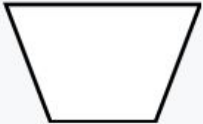


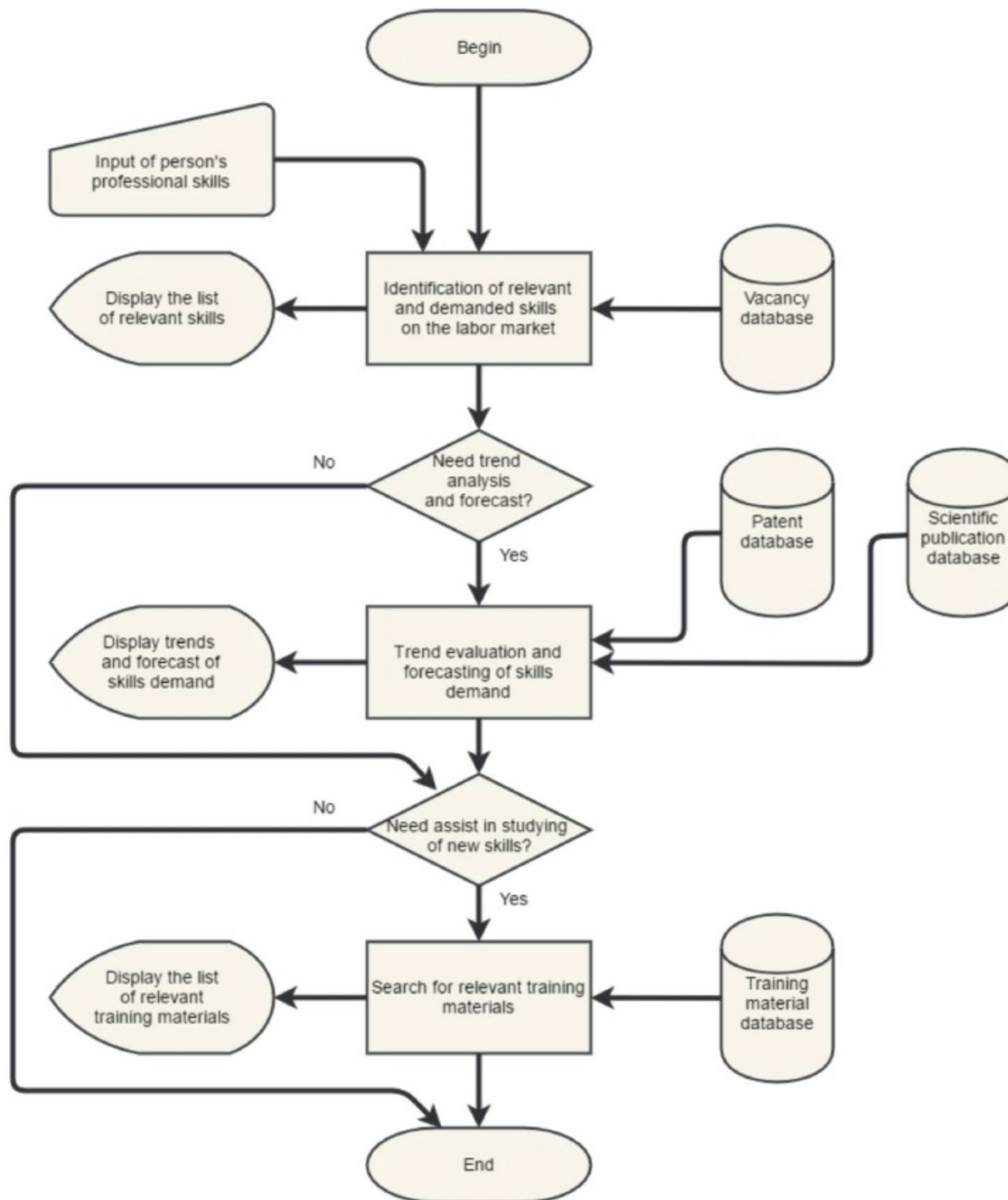
Comment



Predefined process (process is defined elsewhere)

Assignment: Approach (Flowchart)

Shape	Name
	Data File or Database
 	Document
	Manual operation
	Manual input
	Preparation or Initialization



Assignment: Experimental Evaluation

- Evaluation & user experience evaluation
- We expect of the evaluation:
 - > Research Question
 - > Hypothesis
 - > Experimental and Control Condition
 - > Independent Variable
 - > Dependent variable
 - > Subjects/participants
 - > Possible Confounding variables
- User experience evaluation (with questionnaire that will be used to evaluate your system (eg. acceptability))
- Today: Work session!
- Submit Friday 14 Oct

Assignment: Experimental Evaluation (example)

Research Question: *Is it possible to reduce smoking behavior by using a smart app that gives tailored advice to its users?*

Hypothesis: A smart app reduces smoking behavior more than an app that gives only information to its users, and little to no tailored advice.

Control Condition and Experimental Condition:

Half of the test subjects are given an app that provides only information to its users, and little to no tailored advice, while the remaining half is given a smart app that gives tailored advice to its users.

Dependent variable: smoking behavior (e.g. number of weekly cigarettes)

Independent Variable: Intervention (with or without the smart app)

Test Subjects: E.g. Male, have been smoking for more than 10 years.

Possible Confounding Variables: E.g. the interface is not user-friendly,
and therefore the participants stop using the app.

Feedback “Literature & Knowledge Acquisition”

- Feedback in Canvas
- Questions?